PVP 23

Code: 23CS3503

III B.Tech - I Semester - Regular Examinations - NOVEMBER 2025

FORMAL LANGUAGES AND AUTOMATA THEORY
(COMPUTER SCIENCE & ENGINEERING)

Duration: 3 hours Max. Marks: 70

Note: 1. This question paper contains two Parts A and B.
2. Part-A contains 10 short answer questions. Each Question carries 2 Marks.
3. Part-B contains 5 essay questions with an internal choice from each unit. Each Question carries 10 marks.
4. All parts of Question paper must be answered in one place.

BL – Blooms Level CO – Course Outcome

PART – A

		BL	CO
1.a)	Define DFA.	L1	CO1
b)	Why it is important to study Automata Theory for Computer science?	L2	CO1
c)	Relate regular grammars and regular expressions.	L2	CO1
d)	Specify the reason for eliminating useless symbols. How to identify them?	L2	CO2
e)	Write the advantages of parse tree in identifying ambiguity.	L2	CO2
f)	What is Left recursion?	L1	CO2
g)	Define Push Down Automata.	L2	CO2
h)	How to convert the context free grammar to Push Down Automata?	L2	CO2
i)	What do you mean by Instantaneous Description of Turing Machine?	L2	CO3
j)	Define Post Correspondence Problem.	L2	CO4

PART – B

			BL	CO	Max. Marks
UNIT-I					
2	a)	Define NFA. What are the differences between DFA & NFA?	L3	CO1	5 M
	b)	Explain the advantages of Finite State Machine.	L2	CO1	5 M
OR					
3	a)	Design DFA to accept strings with 'c' and 'd' such that number of d's are divisible by 4.	L3	CO2	5 M
	b)	Explain the procedure for constructing minimum state DFA with an example.	L3	CO2	5 M
UNIT-II					
4	a)	Discuss Algebraic Laws for regular expressions.	L2	CO2	5 M
	b)	Construct a DFA for the Regular expression (0+1)* (00+11) (0+1)*	L3	CO2	5 M
OR					
5	What is a regular language? What is relationship between finite automata and regular expression? Explain the process of converting DFA to regular expression.		L2	CO2	10 M
UNIT-III					
6	a)	Find the left most derivation for the string abba in the grammar S→AA, A→aB, B→bB/ε	L3	CO2	5 M

	b)	Explain about Ambiguity in Grammars and Languages with example.	L3	CO2	5 M
OR					
7		State Pumping Lemma for context free languages and explain how the given CFG is not context free. L = $\{a^n b^n c^n \mid n>=1\}$	L4	CO4	10 M
UNIT-IV					
8	a)	Construct PDA for the following language. L = $\{ a^n b^n c^m d^m \mid n>=1, m>=1\}$	L3	CO2	5 M
	b)	Explain the basic structure of PDA with a neat graphical representation.	L2	CO2	5 M
OR					
9	a)	Convert the following grammar to PDA S→0AA , A→0S/1S/0	L3	CO2	5 M
	b)	What is deterministic Push Down Automata? Explain.	L2	CO2	5 M
UNIT-V					
10	a)	Construct a Turing Machine that will accept the Language consists of all palindromes of 0's and 1's.	L3	CO3	5 M
	b)	Explain about types of Turing Machine.	L2	CO3	5 M
OR					
11	a)	Explain about the Universal Turing machine (UTM).	L2	CO3	5 M
	b)	How can the Halting problem of Turing machine be handled? Explain.	L2	CO4	5 M